

Python Syntax Cheat Sheet

Variables & Data Types

```
x = 10    # Integer
y = 3.14  # Float
name = 'Farah'    # String
is_active = True  # Boolean
fruits = ['apple', 'banana', 'cherry']    # List
info = {'name': 'Farah', 'age': 21}    # Dictionary
numbers = (1, 2, 3)    # Tuple
unique_vals = {1, 2, 3}    # Set
```

Conditional Statements

```
if x > 5:    print('x is greater than 5')
elif x == 5:    print('x is 5')
else:    print('x is less than 5')
```

Loops

```
for i in range(5):    print(i)    # 0 to 4
while x > 0:    x -= 1
```

List Comprehension

```
squares = [x**2 for x in range(5)]    # [0, 1, 4, 9, 16]
```

Functions

```
def greet(name):    return f'Hello, {name}!'
print(greet('Farah'))    # Hello, Farah!
```

Lambda Functions

```
square = lambda x: x ** 2    # square(4) -> 16
```

Dictionary Operations

```
info.keys()    # Get all keys
info.values()    # Get all values
info.items()    # Get key-value pairs
```

Exception Handling

```
try:    print(10 / 0)
except ZeroDivisionError:    print('Cannot divide by zero')
```

File Handling

```
with open('file.txt', 'r') as f:    content = f.read()
```

```
# Importing Modules
import math    # math.sqrt(16) -> 4.0
from collections import Counter    # Counter([1,2,2,3])

# List Sorting
nums.sort()    # Sort in-place
sorted(nums, reverse=True)    # Return sorted copy

# Enumerate & Zip
for idx, val in enumerate(fruits):    print(idx, val)
for a, b in zip(list1, list2):    print(a, b)

# Random Module
import random
random.randint(1, 10)    # Random int between 1-10
random.choice(fruits)    # Random element from list

# OS Module
import os
os.getcwd()    # Get current directory
os.listdir()    # List files in directory
```